

ALL-HEX CHT MESHER: VALIDATION RESULTS

Summary

The mesher produces valid, wall-resolved, energy-conserving all-hexahedral meshes for conjugate heat transfer (CHT) in liquid cold plates. It was tested on seven cold plates of two families: four pin-fin arrays and three TPMS lattices. Every mesh was generated automatically from a fluid STL and a solid STL, with no manual repair. Every mesh passes all OpenFOAM quality checks, and the coupled simulations conserve the heater power to within 0.13 %.

Result	Acceptance limit	Achieved (7 plates)
checkMesh verdict	"Mesh OK" in every region	Mesh OK, 14 of 14 regions
Inverted cells	0	0
Max non-orthogonality	$\leq 70^\circ$	49.9–69.5°
Max skewness (OpenFOAM)	≤ 4	3.44–3.56
Mean first-cell y^+	< 1	0.40–0.60
Wall-layer coverage	$\geq 99\%$ of the wall	99.8–100 %
Solid volume vs CAD	within 2 %	within 0.32 %
Fluid volume vs CAD	within 1 %	within 0.20 %
Identical fluid/solid interface faces	–	99.0–100 %
Interface area, fluid vs solid	within 1 %	within 0.06 %
Heater power reaching the coolant	within 1 % of 2000 W	99.991–100.12 %
Interface heat error	within 1 %	0.000–0.13 %

Against SimScale on the same parallel pin-fin case, the outlet water temperature agrees within 0.09 K and the peak heater temperature within 1.3 K. A complete pin-fin plate (6.1–6.6 million cells) is meshed in 6–10 minutes on one GPU node. A TPMS plate (16.4–18.5 million cells) takes 33–111 minutes.

Mesh independence and the sensitivity to the wall-layer transition are still being tested; see Scope, limitations and pending checks.

The mesher

The mesher turns a fluid STL and a solid STL into a two-region OpenFOAM mesh of hexahedral cells ("atoms") and a ready-to-run CHT case. The fluid is meshed first; the solid is then meshed on the fluid's atom grid.

1. **Atoms.** Each body is cut into cubes of one size (0.3 mm for the pin-fin plates, 0.2 mm for the TPMS plates). The solid uses the same lattice as the fluid, offset by whole atoms.
2. **Snapping.** A graph neural network moves the boundary nodes onto the CAD surface. A Jacobian guard keeps every cell valid.
3. **Wall layers (fluid).** Five prism-free hexahedral layers are built at every fluid wall for y^+ below 1: growth ratio 1.3, total height at most $0.4 \times$ atom size.
4. **Export.** OpenFOAM polyMesh for both regions. Every solid interface node takes the position of the fluid node at the same grid point, so both regions share the same interface faces. The solid's other walls are then placed on the CAD surface under a cell-quality guard (scaled Jacobian ≥ 0.2).
5. **Case set-up.** A ready-to-run chtMultiRegionSimpleFoam case is written: inlet, outlet and heater patches, materials, solver settings and run scripts.

Each run writes an accuracy report: the distance of every wall node from the CAD, the mesh volume against the CAD volume, the interface match and the minimum cell quality.

Test cases and simulation setup

Seven copper cold plates cooled by water were meshed and simulated under the same operating point.

Plate	Family	Fluid volume (mm ³)	Solid volume (mm ³)	Wetted area (mm ²)	Atom size
Chevron	Pin-fin	73,680	33,530	39,214	0.3 mm
Cylinder	Pin-fin	64,519	42,671	44,226	0.3 mm
Parallel	Pin-fin	59,070	48,140	48,345	0.3 mm
Square	Pin-fin	67,404	39,806	39,085	0.3 mm
Gyroid	TPMS	50,047	32,329	44,606	0.2 mm
Lidinoid	TPMS	47,395	34,958	59,045	0.2 mm
Neovius	TPMS	49,714	32,659	45,598	0.2 mm

Volumes are those of the CAD (STL); the wetted area is the fluid–solid interface area of the mesh.

Geometry. The pin-fin plates measure $200 \times 70 \times 18$ mm (fluid space $198 \times 68 \times 17$ mm), with two 10.4 mm ports on the top face at $x = 7$ and 193 mm. The TPMS plates are closed plates with a sheet lattice of 0.4 mm walls and two 12 mm ports on the top face at $x = \pm 92.5$ mm. The thin TPMS walls set the atom size: an atom must be at most half the thinnest wall.

Item	Setting
Coolant	Water at 25 °C: $\rho = 997.3 \text{ kg/m}^3$, $\nu = 9.338 \times 10^{-7} \text{ m}^2/\text{s}$, $c_p = 4180 \text{ J}/(\text{kg}\cdot\text{K})$, $Pr = 6.524$; Boussinesq buoyancy
Solid	Copper, $\kappa = 401 \text{ W}/(\text{m}\cdot\text{K})$
Inlet	10 L/min (0.1664 kg/s), 293 K (19.85 °C), turbulence intensity 5 %
Outlet	$p = 10^5 \text{ Pa}$
Heater	2000 W, uniform, on 140×70 mm of the bottom face
Other walls	No-slip; solid outer walls adiabatic
Fluid–solid coupling	mappedWall, turbulentTemperatureCoupledBaffleMixed
Turbulence	k- ω SST, wall treatment valid at any y^+
Schemes	Second order (linearUpwind) for velocity and energy
Solver	chtMultiRegionSimpleFoam, OpenFOAM v2312, 96 MPI processes
Iterations	4000 coupled (flow and heat), then 1000 energy-only iterations with the flow frozen, energy solved to 10^{-10}

The pin-fin flows shed vortices, so pressure drop and temperature oscillate slightly around a steady mean. They are reported with their range over iterations 2500–4000.

Mesh size and meshing time

A pin-fin plate has 3.9 million cells before the wall layers and 6.1–6.6 million with them. A TPMS plate has 10.3 million before and 16.4–18.5 million with them. All cells are hexahedra. Meshing takes 6–10

minutes for a pin-fin plate and 33–111 minutes for a TPMS plate. Wall layers are built only in the fluid, so the solid is the same in both tables.

Without wall layers (snapped atom grid)

Plate	Fluid cells	Fluid nodes	Solid cells	Solid nodes	Total cells	Total nodes
Chevron	2,731,992	2,973,117	1,138,820	1,546,657	3,870,812	4,519,774
Cylinder	2,382,648	2,655,536	1,487,416	1,932,804	3,870,064	4,588,340
Parallel	2,191,512	2,473,296	1,679,300	2,119,260	3,870,812	4,592,556
Square	2,497,252	2,727,417	1,373,560	1,771,579	3,870,812	4,498,996
Gyroid	6,253,639	6,878,356	4,044,824	5,035,212	10,298,463	11,913,568
Lidinoïd	5,920,733	6,764,545	4,375,584	5,567,623	10,296,317	12,332,168
Neovius	6,213,315	6,843,508	4,085,660	5,079,404	10,298,975	11,922,912

With wall layers (the meshes used in all simulations)

Plate	Fluid cells	Fluid nodes	Solid cells	Solid nodes	Total cells	Total nodes
Chevron	5,109,342	5,349,677	1,138,820	1,546,657	6,248,162	6,896,334
Cylinder	5,109,598	5,378,926	1,487,416	1,932,804	6,597,014	7,311,730
Parallel	4,932,222	5,213,766	1,679,300	2,119,260	6,611,522	7,333,026
Square	4,771,032	5,000,087	1,373,560	1,771,579	6,144,592	6,771,666
Gyroid	12,399,179	13,022,966	4,044,824	5,035,212	16,444,003	18,058,178
Lidinoïd	14,169,563	15,011,725	4,375,584	5,567,623	18,545,147	20,579,348
Neovius	12,403,535	13,032,438	4,085,660	5,079,404	16,489,195	18,111,842

The five wall layers add one cell per layer on every fluid wall face and one node per layer on every fluid wall node. That roughly doubles the fluid: +2.3 to 2.7 million cells on a pin-fin plate and +6.1 to 8.3 million on a TPMS plate. The graph neural network snaps 0.45–0.55 million fluid wall nodes on a pin-fin plate and 1.2–1.6 million on a TPMS plate. On the Lidinoïd, layers are left out on 0.165 % of the wall, so its layered node count is an upper bound.

Meshing time in seconds per step, on one cluster GPU node (4 CPU cores, 1 GPU, 64 GB):

Plate	Fluid: atoms	Fluid: snapping	Fluid: wall layers	Fluid: export	Solid: atoms	Solid: snapping	Solid: export	Total
Chevron	16.9	14.2	213.3	143.6	14.9	20.0	68.7	8.2 min
Cylinder	15.6	13.6	328.1	128.9	18.4	20.9	82.1	10.1 min
Parallel	13.7	15.2	99.1	125.2	17.1	23.2	76.1	6.2 min
Square	16.0	14.0	118.8	132.5	15.1	16.8	68.4	6.4 min
Gyroid	101.3	26.2	763.2	491.3	157.2	33.0	412.1	33.1 min
Lidinoid	147.2	28.1	4871.9	730.5	217.9	42.0	618.0	110.9 min
Neovius	101.9	26.4	778.1	505.3	159.7	29.5	423.7	33.7 min

The snapping network itself takes 14–42 s per body. Most of the time goes into the wall layers and the export. The Lidinoid's layer step takes longest because its strongly curved walls need many rounds of the layer-quality guard. The fluid export time includes writing an optional Fluent mesh file.

Mesh quality

All 14 regions pass OpenFOAM checkMesh with "Mesh OK", and no cell is inverted. The worst values are in the thin wall-layer cells of the fluid; the solids sit far inside every limit.

Criterion	What it measures	Acceptance limit	Source
Inverted cells	Cells with negative volume	0	OpenFOAM checkMesh
Min scaled Jacobian	Worst cell shape (1 = cube, ≤ 0 = inverted)	> 0	Standard; the mesher enforces ≥ 0.05 in layers, ≥ 0.2 in the solid
Max non-orthogonality	Angle between face normal and cell-centre line	$\leq 70^\circ$	OpenFOAM primitiveMeshCheck
Max skewness (OpenFOAM)	Face-centre offset from the cell-centre line	≤ 4	OpenFOAM primitiveMeshCheck
Max aspect ratio	Longest to shortest cell	≤ 1000	OpenFOAM

Criterion	What it measures	Acceptance limit	Source
	dimension		primitiveMeshCheck

Fluid regions

Plate	Max non-orthogonality (avg)	Max skewness	Max aspect ratio	Min scaled Jacobian	Cells with J < 0.1	checkMesh
Chevron	53.8° (7.2°)	3.51	47.3	0.075	83	Mesh OK
Cylinder	57.2° (8.1°)	3.44	44.7	0.060	22	Mesh OK
Parallel	50.7° (4.9°)	3.53	39.3	0.060	34	Mesh OK
Square	49.9° (5.3°)	3.56	47.9	0.070	91	Mesh OK
Gyroid	67.0° (11.2°)	3.56	73.6	0.050	15,629	Mesh OK
Lidinoïd	69.5° (13.9°)	3.56	102.9	0.050	40,762	Mesh OK
Neovius	69.4° (10.4°)	3.56	78.5	0.050	17,719	Mesh OK

Solid regions

Plate	Max non-orthogonality (avg)	Max skewness	Max aspect ratio	Min scaled Jacobian	checkMesh
Chevron	29.1°	1.37	2.60	0.238	Mesh OK
Cylinder	28.9°	1.56	2.63	0.203	Mesh OK
Parallel	29.3°	1.22	2.49	0.443	Mesh OK
Square	29.1°	1.35	2.55	0.308	Mesh OK
Gyroid	18.9° (1.7°)	0.79	2.25	0.200	Mesh OK
Lidinoïd	20.4° (2.0°)	0.78	2.29	0.203	Mesh OK
Neovius	22.3° (1.6°)	0.81	2.22	0.205	Mesh OK

- **Skewness** of 3.44–3.56 lies in the 2–4 band that most solvers handle without special treatment.
- **Aspect ratio** of 39–103 is typical for thin wall-layer cells and a tenth of the limit.

- **Non-orthogonality** of 67–70° in the TPMS fluids passes, but with a margin below 1° for Lidinoid and Neovius. The average is only 10–14°, so few faces are affected. The TPMS cases run with two non-orthogonal correctors, the pin-fin cases with one.
- **Low-Jacobian cells** ($J < 0.1$) are all thin first-layer cells on curved walls: 0.25–0.49 % of the layer cells on the TPMS plates and fewer than 100 cells per pin-fin plate.

The TPMS solid values come from checkMesh on the solids before their outer walls were placed on the CAD (a mean node move of 24 μm). The final TPMS solids were checked by the mesher's own export check: minimum scaled Jacobian 0.200–0.205 and no cell below 0.1.

Geometric accuracy and interface match

The meshes reproduce the CAD within every project limit. Fluid and solid share their interface faces, so the two interface areas agree to within 0.06 %.

Criterion	Acceptance limit
Wall deviation (distance of mesh wall nodes from the STL)	Mean $\leq 0.25 \times$ atom size (75 μm at 0.3 mm, 50 μm at 0.2 mm); max $\leq 1 \times$ atom size (300 / 200 μm)
Volume deviation (mesh against STL)	Fluid ≤ 1 %; solid ≤ 2 %
Interface area (fluid side against solid side)	≤ 1 %

Plate	Fluid wall deviation mean / max	Solid outer walls, mean	Fluid volume	Solid volume
Chevron	18.6 / 151.1 μm	1.7 μm	−0.06 %	+0.16 %
Cylinder	20.2 / 178.7 μm	1.7 μm	−0.20 %	+0.32 %
Parallel	19.8 / 175.7 μm	2.2 μm	−0.09 %	+0.14 %
Square	12.8 / 154.2 μm	1.8 μm	+0.02 %	−0.00 %
Gyroid	35.7 / 146.3 μm	1.6 μm	+0.04 %	−0.04 %
Lidinoid	39.2 / 160.2 μm	1.7 μm	+0.12 %	−0.12 %
Neovius	33.9 / 148.0 μm	1.7 μm	+0.04 %	−0.03 %

The mean fluid wall deviation is 4–7 % of the atom size on the pin-fin plates and 17–20 % on the TPMS plates. It is set by sharp edges, which the snapping rounds slightly, and by the curved 0.4 mm TPMS walls, which are only two atoms thick. On the solid's interface the nodes take the fluid's positions, so

they share the fluid's deviation. The solid's outer walls sit on the CAD; on the TPMS solids 99 % of their nodes are within 35 μm , with a maximum of 183 μm at a few nodes where the quality guard limited the move.

Plate	Interface area, fluid side (mm^2)	Interface area, solid side (mm^2)	Difference	Identical interface faces
Chevron	39,214.25	39,213.97	0.00 %	100 %
Cylinder	44,226.26	44,225.94	0.00 %	100 %
Parallel	48,344.85	48,343.27	0.00 %	100 %
Square	39,085.33	39,085.04	0.00 %	100 %
Gyroid	44,605.88	44,630.53	-0.06 %	99.34 %
Lidinoïd	59,045.13	59,073.55	-0.05 %	99.03 %
Neovius	45,597.99	45,591.07	+0.02 %	99.48 %

An identical face is one whose nodes coincide on both sides of the interface. On the TPMS plates the remaining 0.5–1 % are faces where a solid atom has no exact partner in the fluid.

Wall resolution

The mean first-cell y^+ is 0.40–0.60 on all seven plates, so the first cell lies inside the viscous sublayer, as wall-resolved $k-\omega$ SST requires.

Criterion	Acceptance limit	Source
Mean first-cell y^+	< 1 (target about 0.5); < 2 acceptable for engineering accuracy	CFD Handbook, k-ω SST
Layer coverage	≥ 99 % of the wall area	Project
Growth ratio between layers	≤ 1.2 (guideline)	Wolf Dynamics, mesh quality

Plate	Layers	First layer	Last layer	Total height	Growth ratio	Coverage	y^+ mean	y^+ max
Chevron	5	13.3 μm	37.9 μm	120 μm	1.30	100 %	0.49	4.88
Cylinder	5	13.3 μm	37.9 μm	120 μm	1.30	100 %	0.60	5.25

Plate	Layers	First layer	Last layer	Total height	Growth ratio	Coverage	y+ mean	y+ max
Parallel	5	13.3 μm	37.9 μm	120 μm	1.30	100 %	0.40	4.13
Square	5	13.3 μm	37.9 μm	120 μm	1.30	100 %	0.58	4.73
Gyroid	5	8.8 μm	25.3 μm	80 μm	1.30	100 %	0.58	4.47
Lidinoïd	5	8.8 μm	25.3 μm	80 μm	1.30	99.835 %	0.59	30.2
Neovius	5	8.8 μm	25.3 μm	80 μm	1.30	100 %	0.57	4.55

y+ is measured on the fluid side of the interface at the end of the run. Three values do not meet the published guidance:

- **Maximum y+ of 4–5** on a small share of faces, while the mean stays near 0.5. The wall treatment used is valid at any y+, so these faces are still handled correctly.
- **Lidinoïd maximum y+ of 30**, most likely on the 0.165 % of its wall where layers could not be built with valid cells. There the first cell is a full atom.
- **Growth ratio 1.30** against a 1.2 guideline. The larger step is from the last layer to the first core cell: 180 μm after 37.9 μm on the pin-fin plates and 120 μm after 25.3 μm on the TPMS plates, a jump of about 4.7. A test with a smoother transition is pending (see the last section).

Thermal-hydraulic results

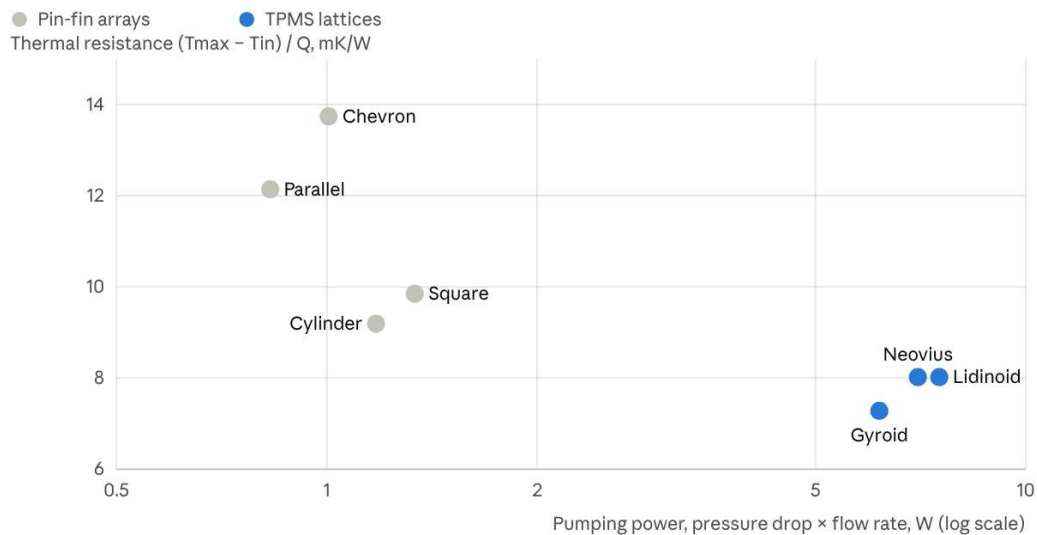
At 10 L/min and 2000 W, the heater stays between 307.6 K and 320.5 K (34.4–47.3 °C), and the outlet water is 2.88 K warmer than the inlet on every plate. Values are at the end of the run; the band is the lowest and highest value over iterations 2500–4000.

Plate	Pressure drop	Pressure drop band	Heater Tmax	Tmax band	Heater Tavg	Outlet bulk T
Chevron	6.03 kPa	5.81–6.10 kPa	320.48 K	319.51–320.82 K	313.56 K	295.875 K
Cylinder	7.06 kPa	6.75–7.45 kPa	311.38 K	310.96–311.60 K	305.25 K	295.875 K
Parallel	4.98 kPa	4.76–5.00 kPa	317.29 K	317.21–317.42 K	310.98 K	295.875 K
Square	8.01 kPa	7.71–8.06 kPa	312.70 K	312.23–312.90 K	307.37 K	295.875 K

Plate	Pressure drop	Pressure drop band	Heater Tmax	Tmax band	Heater Tavg	Outlet bulk T
Gyroid	37.02 kPa	35.96–37.02 kPa	307.55 K	307.38–307.61 K	303.10 K	295.876 K
Lidinoi ¹	45.11 kPa	–	309.05 K	–	301.48 K	295.88 K
Neovius ¹	42.05 kPa	–	309.04 K	–	304.02 K	295.88 K

Thermal resistance against pumping power of the seven cold plates

TPMS lattices: 7.3–8.0 mK/W at 4.6–9.1 times the pumping power of the pin-fin arrays (9.2–13.7 mK/W)



OpenFOAM results at 10 L/min and 2000 W; pumping power = pressure drop × 10 L/min

The TPMS lattices give the lowest thermal resistance and the pin-fin arrays the lowest pumping power. Among the pin-fins, the cylinder array keeps the heater coolest and the parallel array has the lowest pressure drop. The pin-fin flows shed vortices: over iterations 2500–4000 the pressure drop varies by 0.24–0.70 kPa and $T_{\max}$ by 0.21–1.31 K, which is the uncertainty of a steady-state result for these plates.

¹ Lidinoi and Neovius were simulated on meshes whose solid outer walls were not yet placed on the CAD (a mean offset of 24 μm); their final meshes differ only there. On the gyroid the same difference changed $T_{\max}$ by 0.08 K and the pressure drop by less than 0.01 kPa. Their runs on the final meshes are in progress.

Energy conservation

The heater power is conserved through the solid, across the fluid–solid interface and out with the water: 99.991–100.12 % of the 2000 W reaches the coolant, and no heat is lost or created at the interface on the pin-fin plates.

Criterion	Definition	Acceptance limit
Overall balance	Heat carried out by the water ($\dot{m} \cdot c_p \cdot \Delta T$) / heater power	within 1 %
Interface heat error	(heat entering the fluid – heat leaving the solid) / heat leaving the solid	within 1 %
Fluid balance	Heat carried out by the water / heat entering the fluid	within 1 %
Mass balance	Outlet against inlet mass flow	within 0.1 %

Plate	Heat leaving the solid (W)	Heat entering the fluid (W)	Heat to the water (W)	Interface heat error	Overall balance	Fluid balance
Chevron	2000.00	2000.00	1999.91	0.000 %	99.996 %	99.996 %
Cylinder	2000.00	2000.00	1999.92	0.000 %	99.996 %	99.996 %
Parallel	2000.00	2000.00	1999.85	0.000 %	99.993 %	99.993 %
Square	2000.00	2000.00	1999.82	0.000 %	99.991 %	99.991 %
Gyroid	2000.00	2000.72	2000.55	+0.036 %	100.027 %	99.992 %
Lidinoi ¹	2000.00	2002.56	2002.48	+0.128 %	100.124 %	99.996 %
Neovius ¹	2000.00	2000.76	2000.59	+0.038 %	100.030 %	99.992 %

The solid passes 99.9998–100.0000 % of the heater power to the interface in every run. The energy equation is solved to a residual of $1\text{--}2 \times 10^{-10}$ in the last iteration, and the mass flow is conserved to about 10^{-7} relative on the pin-fin plates. On the TPMS plates the small interface error comes from the 0.5–1 % of interface faces that are not identical on both sides.

¹ Simulated on the meshes described in note 1 of the previous section.

Comparison with SimScale

On the same parallel pin-fin plate, the outlet water temperature agrees with SimScale within 0.09 K and the peak heater temperature within 1.3 K. For this comparison the heater covers the whole bottom face (13,101 mm²), at 2000 W, 10 L/min and a 19.85 °C inlet, in both tools.

Quantity	This mesher (OpenFOAM)	SimScale	Difference (ours – SimScale)
Heated area	13,101 mm ²	13,100 mm ²	+1 mm ²
Outlet water temperature, area average	22.68 °C	22.77 °C	–0.09 K
Heater maximum temperature	40.56 °C	41.83 °C	–1.27 K
Heater average temperature	33.77 °C	36.65 °C	–2.88 K
Pressure drop	4.91 kPa	4.25 kPa	+0.66 kPa

- Our pressure drop is the mean over iterations 2500–4000; it varies between 4.62 and 5.23 kPa. Our heater maximum varies between 40.30 and 40.73 °C over the same iterations.
- Our mass-weighted outlet temperature is 22.73 °C, which is exactly what 2000 W heats 10 L/min of water to (+2.88 K). Our energy balance closes at 99.987 % with an interface heat error of 0.000 %.
- SimScale also reports a minimum heater-face temperature of 26.65 °C and channel velocities of 0.80–0.83 m/s near the side walls and 0.29–0.36 m/s in the middle; these were not monitored in our run.

Setting	This mesher (OpenFOAM)	SimScale
Mesh	4.93 M fluid + 1.68 M solid cells, all hexahedral, 0.3 mm	Automatic mesh, "Perfect" setting; cell count not reported
Wall resolution	Mean y+ 0.40 (max 4.13); viscous sublayer resolved	y+ not below 1
Convection scheme (velocity, energy)	Second order	First order
Iterations	4000 coupled + 1000 energy	1000

Setting	This mesher (OpenFOAM)	SimScale
	only	

Both tools put the same heat into the water. The remaining differences point the same way: our run has the higher pressure drop and the cooler heater. That is the direction expected from a boundary layer resolved down to $y^+ 0.4$ with a second-order scheme, which captures the wall friction and heat transfer around the pins more sharply. Which result is closer to a real plate would need a measurement.

Computational cost

From STL to converged result, a pin-fin plate takes about 1.1 hours and a TPMS plate 3–5 hours on the RWTH CLAIX cluster.

Plate	Total cells	Meshing (1 GPU node)	Solver wall time, 5000 iterations (96 cores)	Solver core-hours
Chevron	6.25 M	8.2 min	1.00 h	96
Cylinder	6.60 M	10.1 min	0.98 h	94
Parallel	6.61 M	6.2 min	0.98 h	94
Square	6.14 M	6.4 min	0.96 h	92
Gyroid	16.44 M	33.1 min	3.33 h	320
Lidinoid	18.55 M	110.9 min	3.09 h	296
Neovius	16.49 M	33.7 min	2.68 h	257

- Meshing node: 4 CPU cores, 1 GPU, 64 GB memory. Solver: 96 MPI processes, OpenFOAM v2312.
- The pin-fin and gyroid solver times include one restart from a saved state.
- Writing the CHT case (patches, materials, settings, run scripts) is part of the export and takes